Appendix:

Appendix A presents the complete mathematical formulations for the planning master problem and operation subproblem of the two-stage distributionally robust optimization model. Appendix B provides the proof of the E-C&CG algorithm. Appendix C provides the method to determine the number of typical days.

Appendix A. Decomposition of the two-stage distributionally robust optimization model

1) Planning master problem

$$\min_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{y}^d,Y} \left(C_{\mathrm{GZ}}\left(\boldsymbol{x}\right) + Y \right) \tag{A1}$$

$$Y \geq C_{\mathrm{OP}}^{d}\left(\boldsymbol{y}^{d}, \boldsymbol{p}^{d}\right)\ \forall \boldsymbol{p} \in \Gamma,\ \forall d = 1,2,...,i \tag{A2}$$

$$C_{\mathrm{OP}}^{d} = C_{\mathrm{COAL}}^{d} + C_{\mathrm{LIFELOSS}}^{d} + C_{\mathrm{BIO}}^{d} + C_{\mathrm{NH_3}}^{d} + C_{\mathrm{SHED}}^{d} + C_{\mathrm{LOSSRE}}^{d} + C_{\mathrm{CO2}}^{d} \tag{A3}$$

$$\sum_{h=2}^{H} u_{n,h} \leq 1\ \forall n \tag{A4}$$

$$\sum_{w=1}^{W} u_{w}^{\mathrm{NH_3}} \geq u_{n,3}\ \forall n \tag{A5}$$

$$s_{n,j,t}^{\mathrm{BIO},d} = s_{n,j,t-1}^{\mathrm{BIO},d} + f_{n,j,t}^{\mathrm{BIO,IN},d}\eta_{\mathrm{in}}^{\mathrm{IN,BIO}} - f_{n,j,t}^{\mathrm{BIO,FIRE},d} / \eta_{\mathrm{out}}^{\mathrm{IN,BIO}}\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A6}$$

$$\sum_{h=1}^{H} u_{n,h} S_{h,\min}^{\mathrm{BIO}} \leq s_{n,j,t}^{\mathrm{BIO},d} \leq \sum_{h=1}^{H} u_{n,h} S_{h,\max}^{\mathrm{BIO}}\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A7}$$

$$s_{n,j,0}^{\mathrm{BIO},d} = s_{n,j,T}^{\mathrm{BIO},d} = \sum_{h=1}^{H} u_{n,h} S_{h,0}^{\mathrm{BIO}}\ \forall n,\ \forall j,\ \forall d = 1,2,...,i \tag{A8}$$

$$0 \leq f_{n,j,t}^{\mathrm{BIO,IN},d} \leq F_{n,\max}^{\mathrm{BIO,IN}}\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A9}$$

$$f_{n,j,t}^{\mathrm{BIO,FIRE},d} = \frac{p_{n,j,t}^{\mathrm{BIO},d}}{Q^{\mathrm{BIO}}}\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A10}$$

$$\delta_{n}^{\mathrm{BIO}} = \frac{g_{n,j,t}^{\mathrm{BIO},d}}{Q^{\mathrm{BIO}}} / \left(\frac{g_{n,j,t}^{\mathrm{BIO},d}}{Q^{\mathrm{BIO}}} + \frac{g_{n,j,t}^{\mathrm{COAL},d}}{Q^{\mathrm{COAL}}} \right)\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A11}$$

$$\delta_n^{\mathrm{BIO}} = \sum_{h=1}^{H} \delta_h^{\mathrm{CBIO}} u_{n,h} \ \ \forall n \tag{A12}$$

$$q_{n,j,t}^{\mathrm{MIX,BIO}} = \delta_n^{\mathrm{BIO}} \cdot Q^{\mathrm{BIO}} + \left(1-\delta_n^{\mathrm{BIO}}\right) \cdot Q^{\mathrm{COAL}} \ \ \forall n,\ \forall j,\ \forall t \tag{A13}$$

$$g_{n,j,t}^{\mathrm{BIO,TOTAL},d} = q_{n,j,t}^{\mathrm{MIX,BIO}} \left(\frac{g_{n,j,t}^{\mathrm{BIO},d}}{Q^{\mathrm{BIO}}} + \frac{g_{n,j,t}^{\mathrm{COAL},d}}{Q^{\mathrm{COAL}}} \right) \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A14}$$

$$g_{w,j,t}^{\mathrm{EL},d} = \frac{Q^{\mathrm{H_2}} f_{w,j,t}^{\mathrm{H_2},d}}{\eta_{\mathrm{EL}}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A15}$$

$$\sum_{w=1}^{W} u_w^{\mathrm{NH_3}} G_{w,\min}^{\mathrm{EL}} \le g_{w,j,t}^{\mathrm{EL},d} \le G_{\max}^{\mathrm{EL}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A16}$$

$$g_{w,j,t}^{\mathrm{AS},d} = \frac{f_{w,j,t}^{\mathrm{Air},d} R T_1^{\mathrm{tem}} \ln\left(P_4 / P_1\right)}{M_{\mathrm{Air}} \eta_{\mathrm{N_2,com}}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A17}$$

$$f_{w,j,t}^{\mathrm{Air},d} = \frac{f_{w,j,t}^{\mathrm{N_2},d} \eta_{\mathrm{AS}}}{\pi_{N_2}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A18}$$

$$f_{w,j,t}^{\mathrm{N_2},d} = Zua_{\mathrm{NH_3}}^{\mathrm{N_2}} f_{w,j,t}^{\mathrm{NH_3},d} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A19}$$

$$f_{w,j,t}^{\mathrm{NH_3},d} = f_{w,j,t-1}^{\mathrm{NH_3},d} + r_{w,j,t}^{\mathrm{NH_3},d} \Delta t \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A20}$$

$$f_{w,j,0}^{\mathrm{NH_3},d} = 0 \ \ \forall w,\ \forall j,\ \forall d = 1,2,...,i \tag{A21}$$

$$0 \le f_{w,j,t}^{\mathrm{NH_3},d} \le F_{w,\max}^{\mathrm{NH_3}} u_w^{\mathrm{NH_3}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A22}$$

$$u_w^{\mathrm{NH_3}} R_{w,\min}^{\mathrm{NH_3}} \le r_{w,j,t}^{\mathrm{NH_3},d} \le R_{\max}^{\mathrm{NH_3}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A23}$$

$$g_{w,j,t}^{\mathrm{LOSS,NH_3},d} = Cua^{\mathrm{NH_3}} f_{w,j,t}^{\mathrm{NH_3},d} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A24}$$

$$g_{w,j,t}^{\mathrm{TotalLoss,NH_3},d} = g_{w,j,t}^{\mathrm{EL},d} + g_{w,j,t}^{\mathrm{H_2,COM},d} + g_{w,j,t}^{\mathrm{AS},d} + g_{w,j,t}^{\mathrm{LOSS,NH_3},d} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A25}$$

$$s_{w,j,t}^{\mathrm{F},d} = s_{w,j,t-1}^{\mathrm{F},d} + f_{w,j,t}^{\mathrm{F},d} \eta_{\mathrm{in}}^{\mathrm{F,ST}} - f_{w,j,t}^{\mathrm{F,out},d} / \eta_{\mathrm{out}}^{\mathrm{F,ST}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A26}$$

$$u_w^{\mathrm{NH_3}} S_{w,\min}^{\mathrm{F}} \le s_{w,j,t}^{\mathrm{F},d} \le S_{w,\max}^{\mathrm{F}} u_w^{\mathrm{NH_3}} \ \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A27}$$

$$s_{w,j,0}^{\mathrm{F},d} = s_{w,j,T}^{\mathrm{F},d} = u_w^{\mathrm{NH_3}} S_{w,0}^{\mathrm{F}} \ \ \forall w,\ \forall j,\ \forall d = 1,2,...,i \tag{A28}$$

$$f_{w,j,t}^{\mathrm{H_2,out},d} = Zua_{\mathrm{NH_3}}^{\mathrm{H_2}} f_{w,j,t}^{\mathrm{NH_3},d} \quad \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A29}$$

$$g_{w,j,t}^{\mathrm{H_2,COM},d} = \frac{f_{w,j,t}^{\mathrm{H_2},d} R T_2^{\mathrm{tem}} \ln\left(P_3 / P_2\right)}{M_{\mathrm{H_2}} \eta_{\mathrm{H_2,com}}} \quad \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A30}$$

$$0 \le f_{w,n,j,t}^{\mathrm{NH_3,OUT},d} \le u_w^{\mathrm{NH_3}} F_{w,\max}^{\mathrm{NH_3,OUT}} \quad \forall w,\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A31}$$

$$s_{n,j,t}^{\mathrm{IN,NH_3},d} = s_{n,j,t-1}^{\mathrm{IN,NH_3},d} + \sum_{w=1}^{W} f_{w,n,j,t-\tau_{w,n}}^{\mathrm{NH_3,OUT},d} \eta_{\mathrm{IN}}^{\mathrm{IN,NH_3ST}} - f_{n,j,t}^{\mathrm{NH_3,FIRE},d} / \eta_{\mathrm{OUT}}^{\mathrm{IN,NH_3ST}}$$
$$\forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A32}$$

$$\sum_{h=1}^{H} u_{n,h} S_{h,\min}^{\mathrm{IN,NH_3}} \le s_{n,j,t}^{\mathrm{IN,NH_3},d} \le \sum_{h=1}^{H} u_{n,h} S_{h,\max}^{\mathrm{IN,NH_3}} \quad \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A33}$$

$$s_{n,j,0}^{\mathrm{IN,NH_3},d} = s_{n,j,T}^{\mathrm{IN,NH_3},d} = \sum_{h=1}^{H} u_{n,h} S_{h,0}^{\mathrm{IN,NH_3}} \quad \forall n,\ \forall j,\ \forall d = 1,2,...,i \tag{A34}$$

$$g_{n,j,t}^{\mathrm{NH_3},d} = \frac{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE},d} Q^{\mathrm{NH_3}}}{\Delta t} \quad \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A35}$$

$$\delta_n^{\mathrm{NH_3}} = \frac{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE},d} Q^{\mathrm{NH_3}}}{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE},d} Q^{\mathrm{NH_3}} + e_{n,j,t}^{\mathrm{COAL},d} Q^{\mathrm{COAL}}} \quad \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A36}$$

$$e_{n,j,t}^{\mathrm{COAL},d} m_{\mathrm{coal}} = a_n \left(g_{n,j,t}^{\mathrm{COAL},d}\right)^2 + b_n g_{n,j,t}^{\mathrm{COAL},d} + c_n \quad \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A37}$$

$$\delta_n^{\mathrm{NH_3}} = \sum_{h=1}^{H} u_{n,h} \delta_h^{\mathrm{CNH_3}} \quad \forall n,\ \forall j,\ \forall t \tag{A38}$$

$$q_{n,j,t}^{\mathrm{MIX,NH_3}} = \delta_n^{\mathrm{NH_3}} Q^{\mathrm{NH_3}} + \left(1 - \delta_n^{\mathrm{NH_3}}\right) Q^{\mathrm{COAL}} \quad \forall n,\ \forall j,\ \forall t \tag{A39}$$

$$g_{n,j,t}^{\mathrm{NH_3,TOTAL},d} = q_{n,j,t}^{\mathrm{MIX,NH_3}} \left(\frac{g_{n,j,t}^{\mathrm{NH_3},d}}{Q^{\mathrm{NH_3}}} + \frac{g_{n,j,t}^{\mathrm{COAL},d}}{Q^{\mathrm{COAL}}} \right) \quad \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A40}$$

$$g_{s,n,j,t}^{\mathrm{NET},d} + G_{s,w,j,t}^{\mathrm{RE}} - g_{s,w,j,t}^{\mathrm{LOSSRE},d} - g_{s,\mathrm{w},j,t}^{\mathrm{TatalLoss,NH_3},d} = G_{s,k,j,t}^{\mathrm{LOAD}} - g_{s,k,j,t}^{\mathrm{SHED},d} + \sum_{l}^{L \in \Omega_s} g_{s,l,j,t}^{\mathrm{LINE},d}$$
$$\forall s,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A41}$$

$$g_{s,n,j,t}^{\mathrm{NET},d} = g_{s,n,j,t}^{\mathrm{BIO,TOTAL},d} + g_{s,n,j,t}^{\mathrm{NH_3,TOTAL},d} - g_{s,n,j,t}^{\mathrm{COAL},d} \quad \forall s,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A42}$$

$$G_n^{\min} - \sum_{h=1}^{H} u_{n,h} \Delta G_{n,h} \le g_{n,j,t}^{\mathrm{NET},d} \le G_n^{\max} \ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A43}$$

$$\left| g_{n,j,t}^{\mathrm{NET},d} - g_{n,j,t-1}^{\mathrm{NET},d} \right| \le (r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n,h}) \Delta t \ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A44}$$

$$g_{n,j,t}^{\mathrm{Lifeloss},d} = \max\left\{0, G_n^{\min} - g_{n,j,t}^{\mathrm{NET},d}\right\} \ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A45}$$

$$0 \le g_{w,j,t}^{\mathrm{LOSSRE},d} + g_{w,j,t}^{\mathrm{TotalLoss,NH_3},d} \le G_{w,j,t}^{\mathrm{RE}} \ \forall w,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A46}$$

$$0 \le g_{k,j,t}^{\mathrm{SHED},d} \le G_{s,j,t}^{\mathrm{LOAD}} \ \forall k,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A47}$$

$$g_{l,j,t}^{\mathrm{LINE},d} + (\theta_{a,j,t}^{d} - \theta_{b,j,t}^{d}) / x_l = 0 \ \forall l,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A48}$$

$$\theta_{\min} \le \theta_{s,j,t}^{d} \le \theta_{\max} \ \forall s,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A49}$$

$$\theta_{j,t}^{\mathrm{ref},d} = 0 \ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A50}$$

$$-G_{l,\max}^{\mathrm{LINE}} \le g_{l,j,t}^{\mathrm{LINE},d} \le G_{l,\max}^{\mathrm{LINE}} \ \forall l,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A51}$$

$$sr_{n,j,t}^{\mathrm{UP},d} \le \min\left\{ (r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n.h}) \Delta t,\ G_n^{\max} - g_{n,j,t}^{\mathrm{NET},d} \right\} \ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A52}$$

$$\begin{gathered} sr_{n,j,t}^{\mathrm{DOWN},d} \le \min\left\{ (r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n.h}) \Delta t,\ g_{n,j,t}^{\mathrm{NET},d} - G_n^{\min} + \sum_{h=1}^{H} u_{n,h} \Delta G_{n,h} \right\} \\ \forall n,\ \forall j,\ \forall t,\ \forall d = 1,2,...,i \end{gathered} \tag{A53}$$

$$\begin{cases} \sum_{n=1}^{N} sr_{n,j,t}^{\mathrm{UP},d} \ge R_{j,t}^{\mathrm{UP}} \\ \sum_{n=1}^{N} sr_{n,j,t}^{\mathrm{DOWN},d} \ge R_{j,t}^{\mathrm{DOWN}} \end{cases} \ \forall j,\ \forall t,\ \forall d = 1,2,...,i \tag{A54}$$

2) Operation subproblem

$$\max_{\mathbb{P} \in \Xi} \mathbb{E}_{\mathbb{P}} \left(\min_{\boldsymbol{y}} C_{\mathrm{OP}} \left(\boldsymbol{y} \right) \right) \tag{A55}$$

$$C_{\mathrm{OP}} = C_{\mathrm{COAL}} + C_{\mathrm{LIFELOSS}} + C_{\mathrm{BIO}} + C_{\mathrm{NH_3}} + C_{\mathrm{SHED}} + C_{\mathrm{LOSSRE}} + C_{\mathrm{CO2}} \tag{A56}$$

$$\Xi = \left\{ \mathbf{P} \in \mathbb{P}_{+}^{J} \middle| \begin{array}{l} \sum_{j=1}^{J} \left| p_j - p_j^0 \right| \le \frac{J}{2C} \ln \frac{2J}{1-\delta_1} \\ \max_{1\le j\le J} \left| p_j - p_j^0 \right| \le \frac{1}{2C} \ln \frac{2J}{1-\delta_\infty} \\ \sum_{j=1}^{J} p_j = 1 \\ 0 \le p_j \le 1 \end{array} \right\} \tag{A57}$$

$$s_{n,j,t}^{\mathrm{BIO}} = s_{n,j,t-1}^{\mathrm{BIO}} + f_{n,j,t}^{\mathrm{BIO,IN}} \eta_{\mathrm{in}}^{\mathrm{IN,BIO}} - f_{n,j,t}^{\mathrm{BIO,FIRE}} / \eta_{\mathrm{out}}^{\mathrm{IN,BIO}} \ \ \forall n,\ \forall j,\ \forall t \tag{A58}$$

$$\sum_{h=1}^{H} u_{n,h} S_{h,\min}^{\mathrm{BIO}} \le s_{n,j,t}^{\mathrm{BIO}} \le \sum_{h=1}^{H} u_{n,h} S_{h,\max}^{\mathrm{BIO}} \ \ \forall n,\ \forall j,\ \forall t \tag{A59}$$

$$s_{n,j,0}^{\mathrm{BIO}} = s_{n,j,T}^{\mathrm{BIO}} = \sum_{h=1}^{H} u_{n,h} S_{h,0}^{\mathrm{BIO}} \ \ \forall n,\ \forall j \tag{A60}$$

$$0 \le f_{n,j,t}^{\mathrm{BIO,IN}} \le F_{n,\max}^{\mathrm{BIO,IN}} \ \ \forall n,\ \forall j,\ \forall t \tag{A61}$$

$$f_{n,j,t}^{\mathrm{BIO,FIRE}} = \frac{p_{n,j,t}^{\mathrm{BIO}}}{Q^{\mathrm{BIO}}} \ \ \forall n,\ \forall j,\ \forall t \tag{A62}$$

$$\delta_n^{\mathrm{BIO}} = \frac{g_{n,j,t}^{\mathrm{BIO}}}{Q^{\mathrm{BIO}}} / \left(\frac{g_{n,j,t}^{\mathrm{BIO}}}{Q^{\mathrm{BIO}}} + \frac{g_{n,j,t}^{\mathrm{COAL}}}{Q^{\mathrm{COAL}}} \right) \ \forall n,\ \forall j,\ \forall t \tag{A63}$$

$$\delta_n^{\mathrm{BIO}} = \sum_{h=1}^{H} \delta_h^{\mathrm{CBIO}} u_{n,h} \ \ \forall n \tag{A64}$$

$$q_{n,j,t}^{\mathrm{MIX,BIO}} = \delta_n^{\mathrm{BIO}} \cdot Q^{\mathrm{BIO}} + \left(1 - \delta_n^{\mathrm{BIO}}\right) \cdot Q^{\mathrm{COAL}} \ \ \forall n,\ \forall j,\ \forall t \tag{A65}$$

$$g_{n,j,t}^{\mathrm{BIO,TOTAL}} = q_{n,j,t}^{\mathrm{MIX,BIO}} \left(\frac{g_{n,j,t}^{\mathrm{BIO}}}{Q^{\mathrm{BIO}}} + \frac{g_{n,j,t}^{\mathrm{COAL}}}{Q^{\mathrm{COAL}}} \right) \forall n,\ \forall j,\ \forall t \tag{A66}$$

$$g_{w,j,t}^{\mathrm{EL}} = \frac{Q^{\mathrm{H_2}} f_{w,j,t}^{\mathrm{H_2}}}{\eta_{\mathrm{EL}}} \ \ \forall w,\ \forall j,\ \forall t \tag{A67}$$

$$\sum_{w=1}^{W} u_w^{\mathrm{NH_3}} G_{w,\min}^{\mathrm{EL}} \le g_{w,j,t}^{\mathrm{EL}} \le G_{\max}^{\mathrm{EL}} \ \ \forall w,\ \forall j,\ \forall t \tag{A68}$$

$$g_{w,j,t}^{\mathrm{AS}} = \frac{f_{w,j,t}^{\mathrm{Air}} R T_1^{\mathrm{tem}} \ln\left(P_4 / P_1\right)}{M_{\mathrm{Air}} \eta_{\mathrm{N_2,com}}} \ \forall w,\ \forall j,\ \forall t \tag{A69}$$

$$f_{w,j,t}^{\mathrm{Air}} = \frac{f_{w,j,t}^{\mathrm{N_2}} \eta_{\mathrm{AS}}}{\pi_{N_2}} \ \forall w,\ \forall j,\ \forall t \tag{A70}$$

$$f_{w,j,t}^{\mathrm{N_2}} = Zua_{\mathrm{NH_3}}^{\mathrm{N_2}} f_{w,j,t}^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A71}$$

$$f_{w,j,t}^{\mathrm{NH_3}} = f_{w,j,t-1}^{\mathrm{NH_3}} + r_{w,j,t}^{\mathrm{NH_3}} \Delta t \ \forall w,\ \forall j,\ \forall t \tag{A72}$$

$$f_{w,j,0}^{\mathrm{NH_3}} = 0 \ \forall w,\ \forall j \tag{A73}$$

$$0 \le f_{w,j,t}^{\mathrm{NH_3}} \le F_{w,\max}^{\mathrm{NH_3}} u_w^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A74}$$

$$u_w^{\mathrm{NH_3}} R_{w,\min}^{\mathrm{NH_3}} \le r_{w,j,t}^{\mathrm{NH_3}} \le R_{\max}^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A75}$$

$$g_{w,j,t}^{\mathrm{LOSS,NH_3}} = Cua^{\mathrm{NH_3}} f_{w,j,t}^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A76}$$

$$g_{w,j,t}^{\mathrm{TotalLoss,NH_3}} = g_{w,j,t}^{\mathrm{EL}} + g_{w,j,t}^{\mathrm{H_2,COM}} + g_{w,j,t}^{\mathrm{AS}} + g_{w,j,t}^{\mathrm{LOSS,NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A77}$$

$$s_{w,j,t}^{\mathrm{F}} = s_{w,j,t-1}^{\mathrm{F}} + f_{w,j,t}^{\mathrm{F}} \eta_{\mathrm{in}}^{\mathrm{F,ST}} - f_{w,j,t}^{\mathrm{F,out}} / \eta_{\mathrm{out}}^{\mathrm{F,ST}} \ \forall w,\ \forall j,\ \forall t \tag{A78}$$

$$u_w^{\mathrm{NH_3}} S_{w,\min}^{\mathrm{F}} \le s_{w,j,t}^{\mathrm{F}} \le S_{w,\max}^{\mathrm{F}} u_w^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A79}$$

$$s_{w,j,0}^{\mathrm{F}} = s_{w,j,T}^{\mathrm{F}} = u_w^{\mathrm{NH_3}} S_{w,0}^{\mathrm{F}} \ \forall w,\ \forall j \tag{A80}$$

$$f_{w,j,t}^{\mathrm{H_2,out}} = Zua_{\mathrm{NH_3}}^{\mathrm{H_2}} f_{w,j,t}^{\mathrm{NH_3}} \ \forall w,\ \forall j,\ \forall t \tag{A81}$$

$$g_{w,j,t}^{\mathrm{H_2,COM}} = \frac{f_{w,j,t}^{\mathrm{H_2}} R T_2^{\mathrm{tem}} \ln\left(P_3 / P_2\right)}{M_{\mathrm{H_2}} \eta_{\mathrm{H_2,com}}} \ \forall w,\ \forall j,\ \forall t \tag{A82}$$

$$0 \le f_{w,n,j,t}^{\mathrm{NH_3,OUT}} \le u_w^{\mathrm{NH_3}} F_{w,\max}^{\mathrm{NH_3,OUT}} \ \forall w,\ \forall n,\ \forall j,\ \forall t \tag{A83}$$

$$s_{n,j,t}^{\mathrm{IN,NH_3}} = s_{n,j,t-1}^{\mathrm{IN,NH_3}} + \sum_{w=1}^{W} f_{w,n,j,t-\tau_{w,n}}^{\mathrm{NH_3,OUT}} \eta_{\mathrm{IN}}^{\mathrm{IN,NH_3ST}} - f_{n,j,t}^{\mathrm{NH_3,FIRE}} / \eta_{\mathrm{OUT}}^{\mathrm{IN,NH_3ST}} \ \forall n,\ \forall j,\ \forall t \tag{A84}$$

$$\sum_{h=1}^{H} u_{n,h} S_{h,\min}^{\mathrm{IN,NH_3}} \le s_{n,j,t}^{\mathrm{IN,NH_3}} \le \sum_{h=1}^{H} u_{n,h} S_{h,\max}^{\mathrm{IN,NH_3}} \ \forall n,\ \forall j,\ \forall t \tag{A85}$$

$$s_{n,j,0}^{\mathrm{IN,NH_3}} = s_{n,j,T}^{\mathrm{IN,NH_3}} = \sum_{h=1}^{H} u_{n,h} S_{h,0}^{\mathrm{IN,NH_3}} \ \forall n,\ \forall j \tag{A86}$$

$$g_{n,j,t}^{\mathrm{NH_3}} = \frac{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE}} Q^{\mathrm{NH_3}}}{\Delta t} \ \forall n,\ \forall j,\ \forall t \tag{A87}$$

$$\delta_n^{\mathrm{NH_3}} = \frac{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE}} Q^{\mathrm{NH_3}}}{\eta_{\mathrm{AirNH_3}} f_{n,j,t}^{\mathrm{NH_3,FIRE}} Q^{\mathrm{NH_3}} + e_{n,j,t}^{\mathrm{COAL}} Q^{\mathrm{COAL}}} \ \forall n,\ \forall j,\ \forall t \tag{A88}$$

$$e_{n,j,t}^{\mathrm{COAL}} m_{\mathrm{coal}} = a_n \left(g_{n,j,t}^{\mathrm{COAL}} \right)^2 + b_n g_{n,j,t}^{\mathrm{COAL}} + c_n \ \forall n,\ \forall j,\ \forall t \tag{A89}$$

$$\delta_n^{\mathrm{NH_3}} = \sum_{h=1}^{H} u_{n,h} \delta_h^{\mathrm{CNH_3}} \ \forall n,\ \forall j,\ \forall t \tag{A90}$$

$$q_{n,j,t}^{\mathrm{MIX,NH_3}} = \delta_n^{\mathrm{NH_3}} Q^{\mathrm{NH_3}} + \left(1 - \delta_n^{\mathrm{NH_3}}\right) Q^{\mathrm{COAL}} \ \forall n,\ \forall j,\ \forall t \tag{A91}$$

$$g_{n,j,t}^{\mathrm{NH_3,TOTAL}} = q_{n,j,t}^{\mathrm{MIX,NH_3}} \left(\frac{g_{n,j,t}^{\mathrm{NH_3}}}{Q^{\mathrm{NH_3}}} + \frac{g_{n,j,t}^{\mathrm{COAL}}}{Q^{\mathrm{COAL}}} \right) \ \forall n,\ \forall j,\ \forall t \tag{A92}$$

$$g_{s,n,j,t}^{\mathrm{NET}} + G_{s,w,j,t}^{\mathrm{RE}} - g_{s,w,j,t}^{\mathrm{LOSSRE}} - g_{s,\mathrm{w},j,t}^{\mathrm{TatalLoss,NH_3}} = G_{s,k,j,t}^{\mathrm{LOAD}} - g_{s,k,j,t}^{\mathrm{SHED}} + \sum_{l}^{L \in \Omega_s} g_{s,l,j,t}^{\mathrm{LINE}} \ \forall s,\ \forall j,\ \forall t \tag{A93}$$

$$g_{s,n,j,t}^{\mathrm{NET}} = g_{s,n,j,t}^{\mathrm{BIO,TOTAL}} + g_{s,n,j,t}^{\mathrm{NH_3,TOTAL}} - g_{s,n,j,t}^{\mathrm{COAL}} \ \forall s,\ \forall j,\ \forall t \tag{A94}$$

$$G_n^{\min} - \sum_{h=1}^{H} u_{n,h} \Delta G_{n,h} \le g_{n,j,t}^{\mathrm{NET}} \le G_n^{\max} \ \forall n,\ \forall j,\ \forall t \tag{A95}$$

$$\left| g_{n,j,t}^{\mathrm{NET}} - g_{n,j,t-1}^{\mathrm{NET}} \right| \le \left(r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n,h}\right) \Delta t \ \forall n,\ \forall j,\ \forall t \tag{A96}$$

$$g_{n,j,t}^{\mathrm{Lifeloss}} = \max\left\{0, G_n^{\min} - g_{n,j,t}^{\mathrm{NET}}\right\} \ \forall n,\ \forall j,\ \forall t \tag{A97}$$

$$0 \le g_{w,j,t}^{\mathrm{LOSSRE}} + g_{w,j,t}^{\mathrm{TotalLoss,NH_3}} \le G_{w,j,t}^{\mathrm{RE}} \ \forall w,\ \forall j,\ \forall t \tag{A98}$$

$$0 \le g_{k,j,t}^{\mathrm{SHED}} \le G_{s,j,t}^{\mathrm{LOAD}} \ \forall k,\ \forall j,\ \forall t \tag{A99}$$

$$g_{l,j,t}^{\mathrm{LINE}} + (\theta_{a,j,t} - \theta_{b,j,t}) / x_l = 0 \ \forall l,\ \forall j,\ \forall t \tag{A100}$$

$$\theta_{\min} \le \theta_{s,j,t} \le \theta_{\max} \ \forall s,\ \forall j,\ \forall t \tag{A101}$$

$$\theta_{j,t}^{\text{ref}} = 0 \ \forall j,\ \forall t \tag{A102}$$

$$-G_{l,\max}^{\text{LINE}} \le g_{l,j,t}^{\text{LINE}} \le G_{l,\max}^{\text{LINE}} \ \forall l,\ \forall j,\ \forall t \tag{A103}$$

$$sr_{n,j,t}^{\text{UP}} \le \min\left\{ (r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n.h})\Delta t,\ G_n^{\max} - g_{n,j,t}^{\text{NET}} \right\} \ \forall n,\ \forall j,\ \forall t \tag{A104}$$

$$sr_{n,j,t}^{\text{DOWN}} \le \min\left\{ (r_n^{\max} + \sum_{h=1}^{H} u_{n,h} \Delta r_{n.h})\Delta t,\ g_{n,j,t}^{\text{NET}} - G_n^{\min} + \sum_{h=1}^{H} u_{n,h} \Delta G_{n,h} \right\} \ \forall n,\ \forall j,\ \forall t \tag{A105}$$

$$\begin{cases} \sum_{n=1}^{N} sr_{n,j,t}^{\text{UP}} \ge R_{j,t}^{\text{UP}} \\ \sum_{n=1}^{N} sr_{n,j,t}^{\text{DOWN}} \ge R_{j,t}^{\text{DOWN}} \end{cases} \forall j,\ \forall t \tag{A106}$$

Appendix B. Proof of the E-C&CG algorithm

Lemma 1. For each iteration *i*, the lower bound LB^i of the PMP satisfies $LB^i \le \xi$.

$$\xi = \min_{\boldsymbol{x}} \left\{ \mathbf{c}^{\text{T}} \boldsymbol{x} + \sup_{\mathbb{P} \in \Xi} \mathbb{E}_{\mathbb{P}} (\min_{\boldsymbol{y}} \mathbf{d}^{\text{T}} \boldsymbol{y}) \right\} \tag{B1}$$

Proof. Because $\Gamma \subseteq \Xi$, the PMP is a relaxation of (B1). Hence, the optimal value of the PMP $\text{opt}(\text{PMP}) \le \xi$. The value LB^i produced by the solver is a global lower bound for PMP (i.e., $LB^i \le \text{opt}(\text{PMP})$). Thus $LB^i \le \xi$. □

Lemma 2. The global upper bound $\bar{U}$ maintained by E-C&CG algorithm is non-increasing and satisfies $\bar{U} \ge \xi$ after each iteration *i*.

$$\psi^i = \mathbf{c}^{\text{T}} \boldsymbol{x}^i + \sup_{\mathbb{P} \in \Xi} \mathbb{E}_{\mathbb{P}} (\min_{\boldsymbol{y}} \mathbf{d}^{\text{T}} \boldsymbol{y}) \tag{B2}$$

Proof. The value ψ^i indicated in (B2) is the exact total cost of the feasible first-stage decision $\boldsymbol{x}^i$ under the worst-case PDF; therefore $\psi^i \ge \xi$. Because the E-C&CG updates $\bar{U} = \min\{\bar{U},\ \psi^i\}$ after each iteration *i*, $\bar{U}$ never increases and remains no smaller than ξ. □

Theorem 1. Assume that the ambiguity set Ξ is defined as a polytope with finitely many extreme points and that the subproblem can be solved exactly. Then the E-C&CG algorithm terminates after a finite iteration. Upon termination, the returned solution is ε-optimal, as shown in (B3).

$$\frac{\bar{U}-\xi}{\bar{U}}<\varepsilon \tag{B3}$$

Proof 1. Proof of the finite termination.

First, we suppose that the E-C&CG algorithm does not stop. Let Γ^i and $\varepsilon_{\mathrm{MP}}^i$ denote the scenario set and PMP gap parameter at the beginning of iteration i, respectively.

Because Ξ is a polytope, the set of its extreme points Ξ_{ext} is finite. Every worst-case PDF $\boldsymbol{p}^i$ returned by the OSP can be chosen as an extreme point. Hence, at most $\left|\Xi_{\mathrm{ext}}\right|$ distinct probability vectors can be added to Γ throughout the E-C&CG algorithm. Consequently, there exists an iteration index I^* such that for all $i \geq I^*$, $\Gamma^i=\Gamma$ remains constant.

Since Γ is fixed, for all $i \geq I^*$ the OSP's solution satisfies $\boldsymbol{p}^i \in \Gamma$. Otherwise, a new PDF vector would be added, contradicting the constancy of Γ.

Then, for $i \geq I^*$, the PMP is a fixed optimization problem. The optimal value $\xi_\Gamma \leq \xi$. The solver returns a feasible solution at each iteration, giving a sequence $\left\{UB^i\right\}$ of upper bounds on ξ_Γ. Since the problem is the same and the solver's incumbent never deteriorates, $\left\{UB^i\right\}$ is non-increasing and bounded below by ξ_Γ. Thus $\left\{UB^i\right\}$ will converge, which implies $UB^i-UB^{i-1} \rightarrow 0$. Therefore, the condition $\left(\bar{U}-UB^i\right)/\bar{U}<\tilde{\varepsilon}$ or $UB^i=UB^{i-1}$ will be satisfied infinitely often. The contraction iteration condition is satisfied infinitely many times, and the algorithm repeatedly applies $\varepsilon_{\mathrm{MP}}^* \leftarrow \alpha \varepsilon_{\mathrm{MP}}^*$ with $\alpha \in (0,1)$, resulting $\varepsilon_{\mathrm{MP}}^* \rightarrow 0$.

Solving a fixed optimization problem with a gap tolerance $\varepsilon_{\mathrm{MP}}^*$ that tends to zero forces the solver to produce bounds that converge to the true optimal value, as shown in (B4).

$$\lim_{i\to\infty} LB^i = \lim_{i\to\infty} UB^i = \xi_\Gamma \tag{B4}$$

Finally, for $i \ge I^*$, because $\boldsymbol{p}^i \in \Gamma$, the PMP solution $\left(\boldsymbol{x}^i, Y, \left\{\boldsymbol{y}_j^i\right\}\right)$ satisfies $Y \ge \sum_{j=1}^{J} p_j^i \mathbf{d}^\top \boldsymbol{y}_j^i$ for some feasible recourse vectors $\boldsymbol{y}_j^i$.

$$UB^i = \boldsymbol{c}^\mathrm{T}\boldsymbol{x}^i + Y \ge \boldsymbol{c}^\mathrm{T}\boldsymbol{x}^i + \sum_{j=1}^{J} p_j^i \mathbf{d}^\top \boldsymbol{y}_j^i = \psi^i \ge \xi \ge \xi_\Gamma \tag{B5}$$

Combining (B4) and (B5) yields $\lim_{i\to\infty}\psi^i = \xi_\Gamma$ by the squeeze theorem. The global upper bound $\bar{U}$ is defined as the minimum of all ψ^i encountered so far. Therefore, $\bar{U}$ is non-increasing and bounded below by ξ_Γ, as shown in (B6).

$$\lim_{i\to\infty} \bar{U} = \xi_\Gamma \tag{B6}$$

Consequently, we can obtain Eq. (B7).

$$\lim_{i\to\infty} \frac{\bar{U} - LB^i}{\bar{U}} = \frac{\xi_\Gamma - \xi_\Gamma}{\xi_\Gamma} = 0 \tag{B7}$$

Since the E-C&CG algorithm would terminate as soon as this ratio drops below ε, it must eventually stop. This contradicts the assumption of an infinite iteration in the beginning, proving finite termination. □

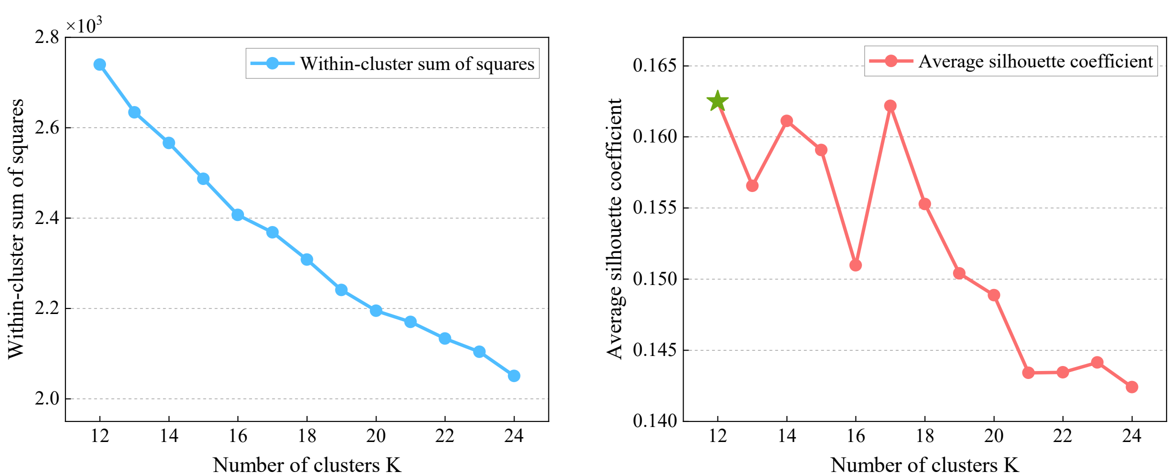
Proof 2. Proof of the ε-optimality.

Let I be the iteration at which the E-C&CG algorithm stops. Then, the termination condition gives $\left(\bar{U} - LB^I\right)/\bar{U} \le \varepsilon$. According to Lemma 1 and Lemma 2, we know that $LB^I \le \xi \le \bar{U}$. Therefore, we can obtain $\frac{\bar{U}-\xi}{\bar{U}} \le \frac{\bar{U}-LB^I}{\bar{U}} \le \varepsilon$, which is exactly the definition of an ε-optimal solution. □

***Remark 3**:* The proof demonstrates that the adaptive reduction of $\varepsilon^{*}_{\mathrm{MP}}$ when the upper bound stagnates is theoretically safe. It preserves the validity of the lower bound LB^{i} and does not impair the finite termination property. On the contrary, it ensures that when the set of scenarios Γ no longer expands, the master problem is eventually solved with sufficient accuracy, guaranteeing convergence in a finite number of iterations. In practice, this heuristic avoids spending excessive time on high-precision solutions of the master problem while the upper bound is still decreasing rapidly, thereby accelerating the overall solution process without sacrificing any optimality guarantee.

Appendix C. The method to determine the number of typical days

Both the Elbow Method and Silhouette Coefficient were utilized to comprehensively evaluate the value of K in the k-means method, as shown in Fig. 18. The results indicate that the highest Silhouette Coefficient (0.1625) was achieved at K = 12. The sum of squares curve showed a continuous downward trend, with no reliable turning point observed. Therefore, we selected 12 to be the value of K based on the evaluation results of both methods.



(a) Results versus Elbow Method (b) Results versus Silhouette Coefficient

Fig. 18. Comparison of different clusters K.